MC connectivity with 100 micrometre simulation steps

28 October 2026

Input: `events/0nubb_100um.next.h5`, converted and verified using the existing Julia hit-table converter. Compare the first 100 catalogue events of this sample with the first 100 of the original double-beta sample. Samples are compared as separate subsets; event IDs do not establish matching trajectories. The 100 micrometre generation step is a user-provided simulation input.

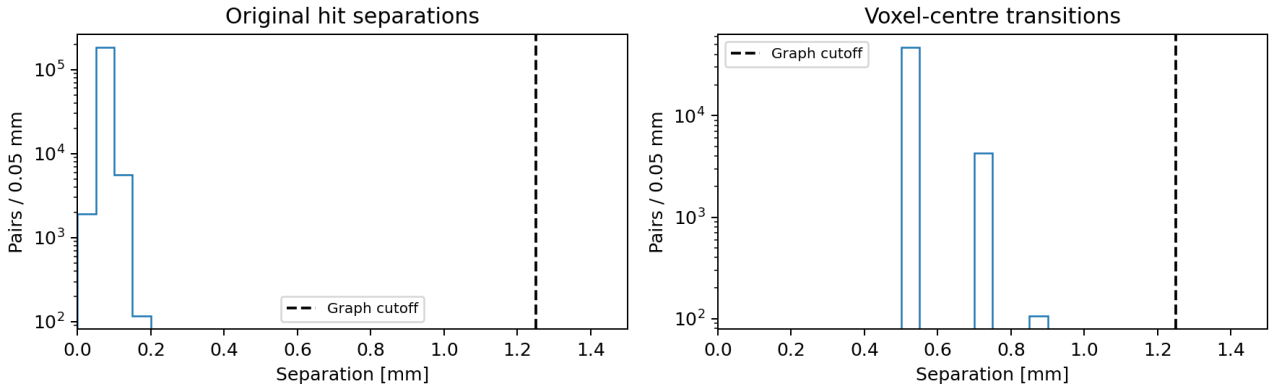
KDT radius graphs use 0.5 mm voxels, source coordinates, all particles, and Float32 calculations. No diffusion or fiducial selection is applied. All connected components, including isolated nodes, are counted.

Radius [mm]	Threshold [keV]	Original: single /100	100 μm: single /100
1.25	1	1	54
1.25	0	1	55
2	1	45	57

Primary hits are sorted separately by (MC time, hit ID). The two orderings agree and there are no equal-time pairs in this subset. Distances are spatial chords between stored deposits, not G4 path lengths. Voxel-transition statistics exclude successive hits in the same cell and precede energy thresholding.

Positions	Pairs	Median [mm]	P99 [mm]	Maximum [mm]
Original hits	191087	0.0994	0.1000	0.2000
Voxel centres	51386	0.5000	0.7071	0.8660

200 of 200 primary-electron voxel clouds form one connected component at a radius of 1.25 mm with no energy threshold. Every measured consecutive primary voxel transition lies within that radius. Thus the step-induced primary fragmentation seen in the original sample is absent here. Remaining event components require inspection of secondary deposits and threshold effects; their count alone is not a background classification.



Numerical directory:
`results/event_analysis/mc_graph_diagnostic/`
Files: `graph_100um.h5`, `hit_spacing_100um.h5`, and `results.h5`.